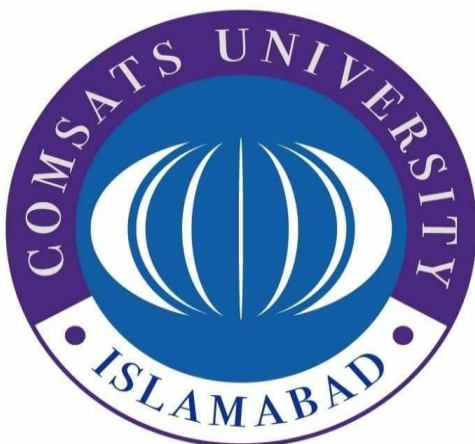


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DATA VISUALIZATION TECHNIQUES

(LAB)

LAB TERMINAL REPORT

NAME:	NAYAB KHALID NOOR UL HUDA BATOOL
REGISTRATION NO.:	SP23-BDS-039 SP23-BDS-040
CLASS:	BSDS-7A
INSTRUCTOR'S NAME:	DR. SHENEELA NAZ
DATE OF SUBMISSION:	19TH JUNE, 2026

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LAB TERMINAL PROJECT

Project Title: Netflix Content Analytics Dashboard

Introduction

The Netflix Content Analytics Dashboard is a comprehensive, interactive data visualization system designed to explore the expansive content library of the world's leading streaming service. The objective of this project is to apply advanced data visualization techniques, specifically using the D3.js library, to uncover meaningful patterns and trends within Netflix's catalog.

We selected the Netflix dataset because it offers a rich blend of categorical, temporal, geographic, and quantitative data. This multi-dimensional complexity presents an excellent opportunity to implement diverse visualization techniques—such as choropleth maps, zoomable treemaps, and focus+context timelines—in a cohesive, cross-filtered environment.

Objectives

The main objectives of this project are:

- Visualize 8,800+ Netflix titles using D3.js
- Build 5 interactive charts (timeline, map, treemap, bar chart, histogram)
- Implement cross-filtering across all visualizations
- Apply brush & zoom for focused data exploration
- Show content trends by year, country, genre, and rating
- Use Netflix-inspired dark UI with glassmorphism design
- Provide built-in report & presentation slides
- Demonstrate core Data Visualization techniques for academic purposes

A guide to our project:

To run the Netflix Analytics Dashboard, open your terminal, navigate to the project folder, and run the command ``python -m http.server 8000``. Once the server is running, open your browser and go to ``http://localhost:8000``.

You will be greeted with an upload screen — simply upload the ``netflix_titles.csv`` file and the dashboard will load automatically.

The dashboard presents seven interactive visualizations: a Content Growth Timeline showing how Netflix's library expanded over the years, a Global Content Map highlighting production by country, a Genre Landscape treemap, a Content Ratings chart, a Duration Distribution histogram,

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a Seasonality chart showing content additions by month, and a Top 10 Directors lollipop chart. On the left sidebar, you can apply filters by year range, content type (Movies or TV Shows), country, rating, and genre — all charts update instantly in real-time when any filter is applied.

Methodology

The project follows a structured data analysis pipeline:

1. Dataset Overview

The primary data source for this project is the 'Netflix Movies and TV Shows' dataset, sourced from Kaggle (**netflix_titles.csv**).

- **Total Records:** 8,807 titles
- **Total Columns:** 12
- **Content Types:** Movies and TV Shows
- **Release Year Range:** 1925 – 2021

Columns Included:

1. Show ID
2. Type (Movie / TV Show)
3. Title
4. Director
5. Cast
6. Country
7. Date Added to Netflix
8. Release Year
9. Rating
10. Duration
11. Genre/Category (listed_in)
12. Description

2. Data Preprocessing

Before visualization, extensive data cleaning and transformation were performed within the data-loader.js module to ensure accurate rendering:

- **Missing Values:** Features like director and country were handled gracefully. Records missing critical temporal data were excluded.
- **Temporal Parsing:** The string format was parsed into JavaScript Date objects to extract structured year_added and month_added dimensions.

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- **Duration Normalization:** The mixed-format duration column was split. Movies were converted to integer minutes, while TV Shows were converted to integer season counts.
- **Multi-value Fields:** Columns containing comma-separated lists were split into arrays. For geographic mapping, the first country in the list was designated as the `primary_country`.

3. Exploratory Data Analysis (EDA)

Our initial exploration revealed several significant trends:

- **Content Split:** Movies dominate the platform, comprising 69.6% of the catalog compared to 30.4% for TV Shows.
- **Growth Trajectory:** Content additions saw exponential growth beginning in 2015, peaking in 2019 with over 2,000 titles added.
- **Geographic Diversity:** The United States is the largest producer (36.5%), followed by India (11.4%) and the UK (7.1%).
- **Rating Distribution:** The platform heavily favors mature content, with TV-MA and TV-14 accounting for over 60% of the entire library.
- **Monthly Patterns:** Content additions are not uniformly distributed across the year; certain months — particularly July and December — show consistently higher addition counts, suggesting deliberate seasonal release strategies.
- **Director Dominance:** A small group of prolific directors account for a disproportionately large share of Netflix's catalog, with the top 10 directors collectively contributing hundreds of titles.

4. Visualization Design & Justification

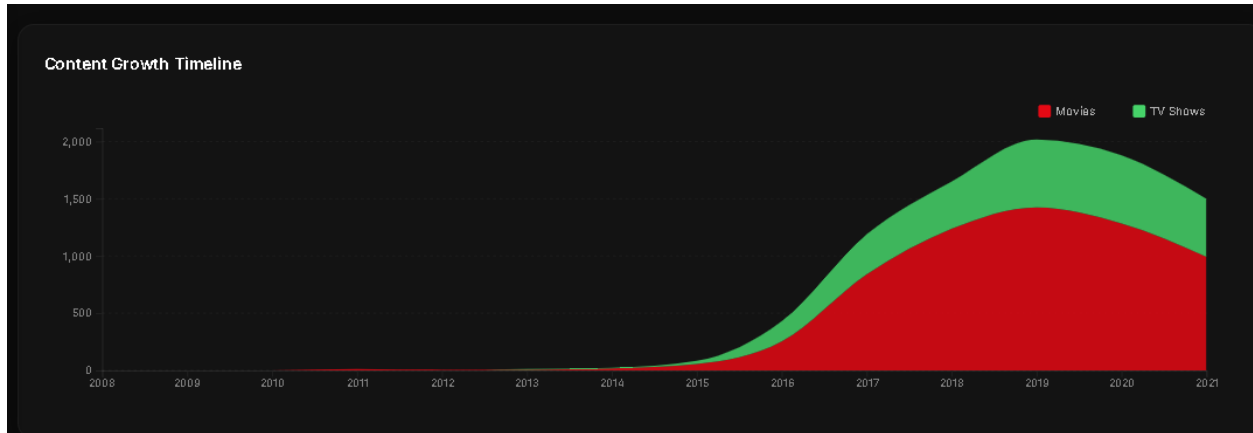
Our dashboard implements seven distinct visualizations, each carefully designed following Tamara Munzner's Nested Model of Visualization Design.

4.1 Content Growth Timeline (Stacked Area Chart)

Task: Analyze the temporal distribution of content additions while comparing Movies and TV Shows.

Design: We utilized a stacked area chart. The horizontal position encodes time, while vertical position and area encode the count. Color hue distinguishes content type (Netflix Red for Movies, Green for TV Shows). A brushable context area implements Shneiderman's 'overview first, zoom and filter' paradigm.

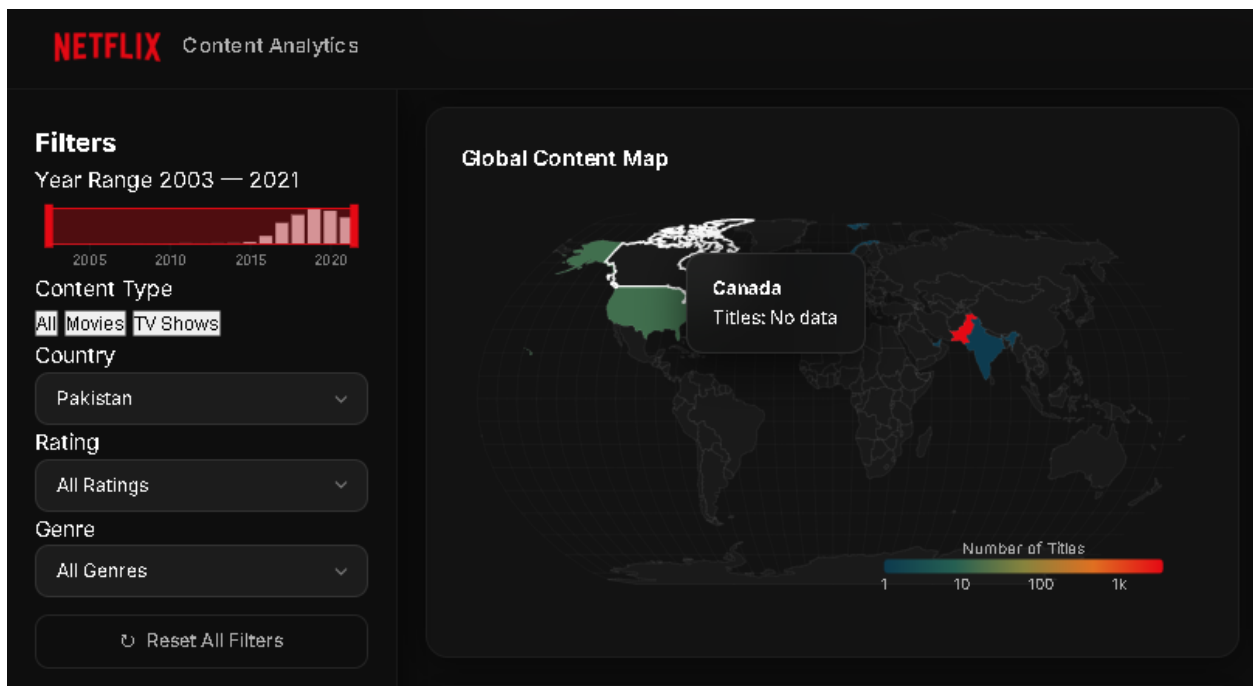
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4.2 Global Content Map (Choropleth)

Task: Identify geographic patterns and production hubs globally.

Design: A choropleth map was chosen to naturally map geographic regions to spatial positions. We utilized a sequential color scale to encode the title count. The interactive tooltips and zoom capabilities mitigate the limitations of area size.

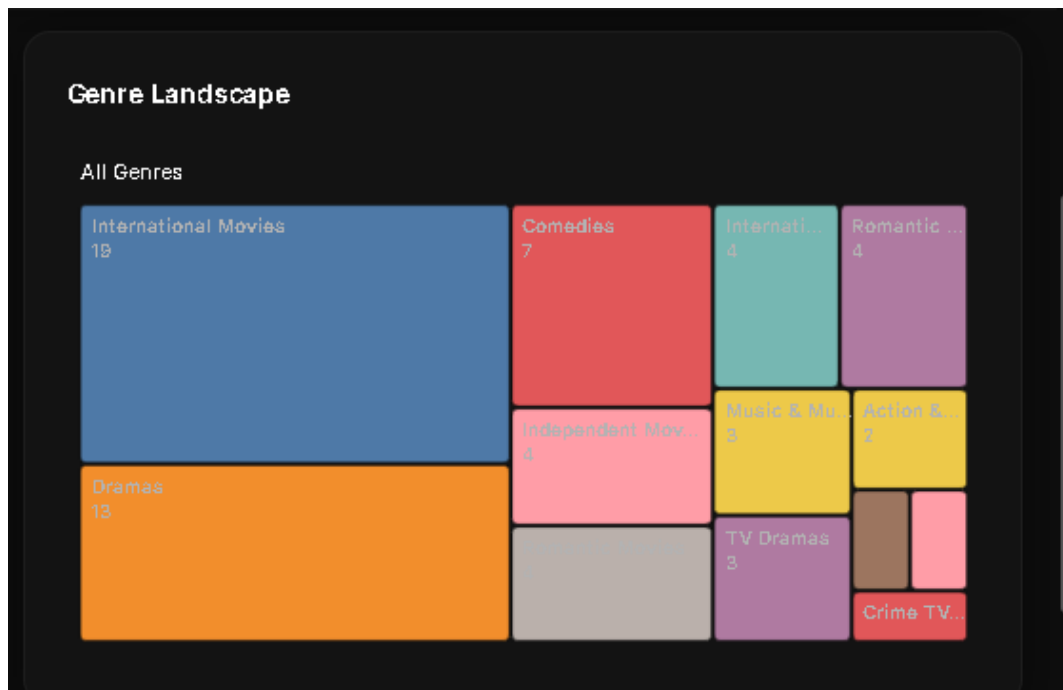


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4.3 Genre Landscape (Zoomable Treemap)

Task: Explore the hierarchical composition of genres.

Design: The treemap efficiently utilizes space to display part-to-whole relationships. Area size strictly encodes the title count, while color hue differentiates major genre categories.

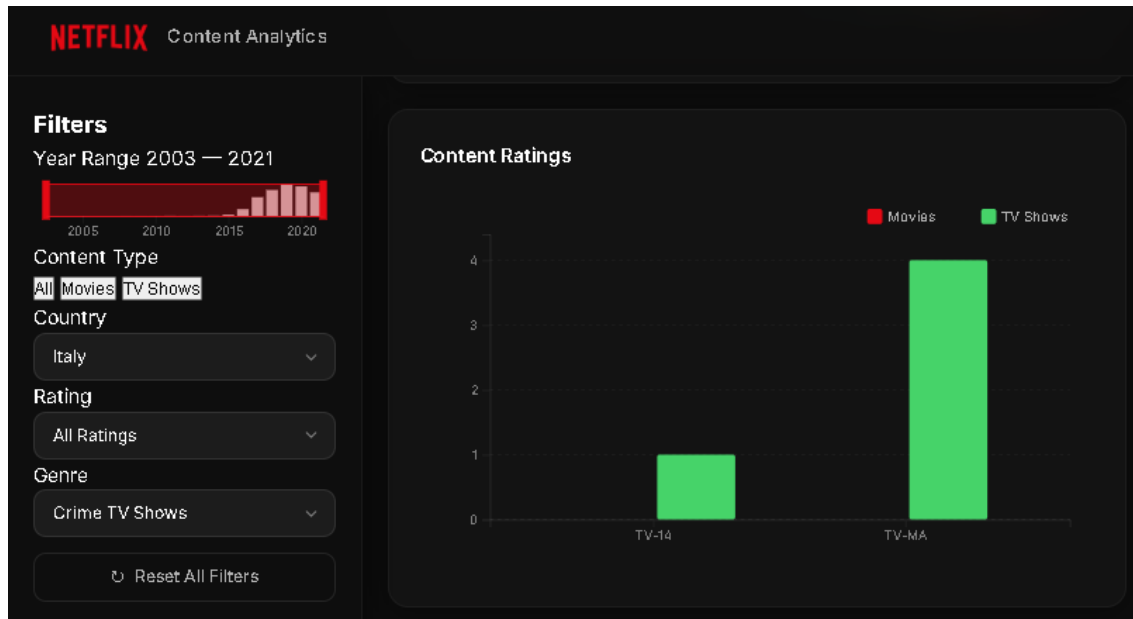


4.4 Content Ratings (Grouped Bar Chart)

Task: Compare rating distributions across the two distinct content types.

Design: A grouped bar chart enables direct, precise comparisons. Vertical length encodes the title count, while the horizontal axis is ordered logically by maturity level rather than alphabetically.

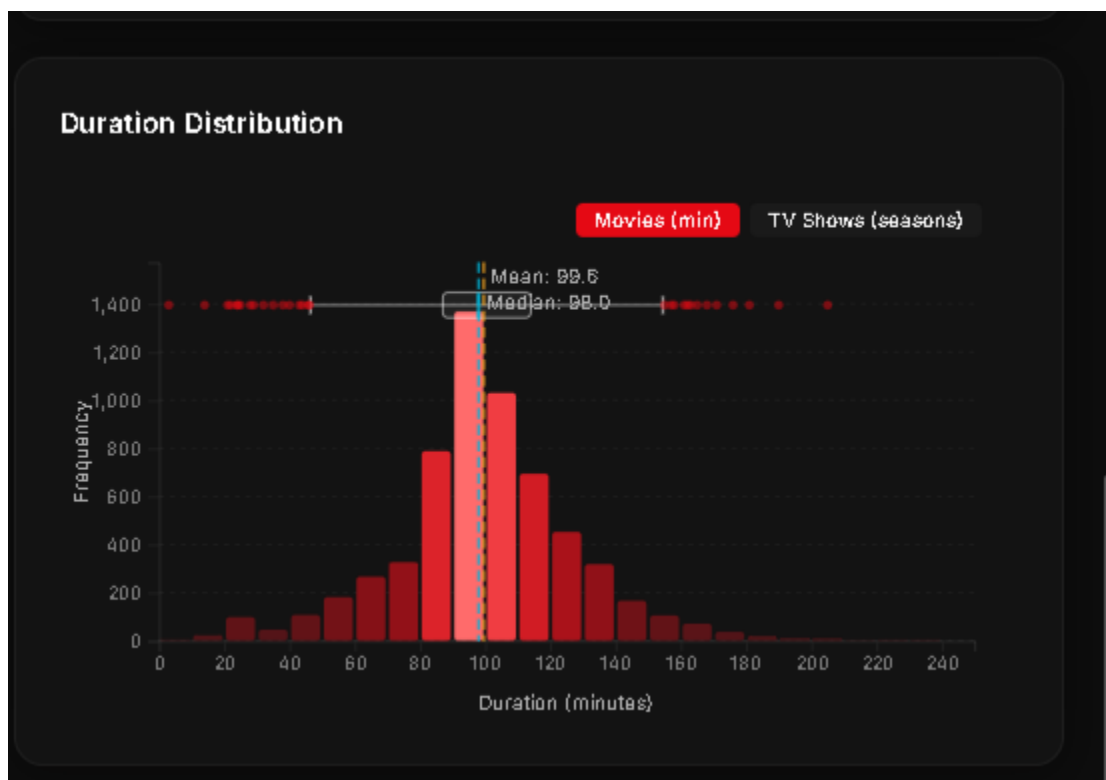
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4.5 Duration Distribution (Histogram + Box Plot)

Task: Understand the statistical distribution of content lengths.

Design: A dual-mode histogram displays frequencies (10-minute bins for movies, discrete bars for seasons). A statistical box plot overlay visualizes the median, interquartile range, whiskers, and outlier data points dynamically.

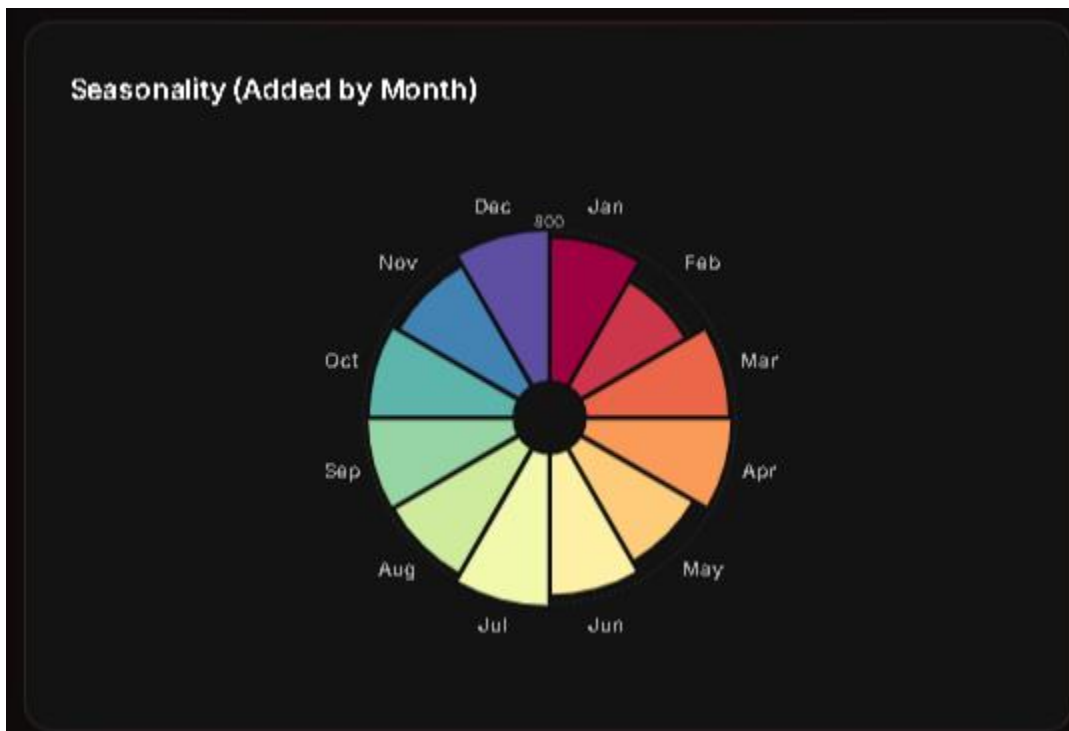


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4.6 Seasonality — Content Added by Month (Radial Bar Chart)

Task: Reveal monthly patterns in Netflix's content addition strategy across the calendar year.

Design: A radial (polar) bar chart was selected to naturally represent the cyclical, month-over-month nature of the data. Angular position encodes the month, while the radial length of each arc encodes the title count. A sequential spectral color scale is applied across the 12 months to provide immediate visual distinction between segments. Concentric grid circles with count labels provide a reference frame for magnitude comparison. Interactive tooltips display the exact title count on hover, and smooth arc transitions are applied when filters update the underlying data.

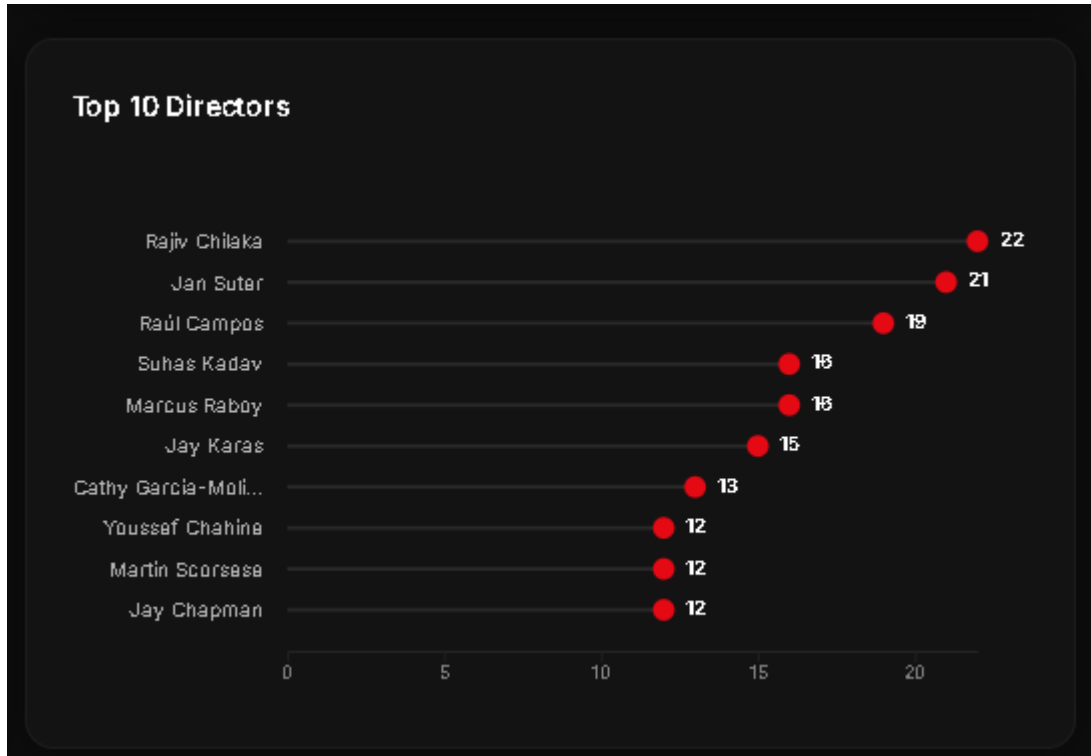


4.7 Top 10 Directors (Lollipop Chart)

Task: Identify the most prolific directors in Netflix's catalog and compare their output.

Design: A horizontal lollipop chart was chosen over a standard bar chart to reduce visual clutter while maintaining precise readability. Director names are encoded on the vertical (band) axis, while the horizontal position of each circle encodes the title count via a linear scale. Animated lines grow from zero on load, reinforcing the magnitude of each director's contribution. Numeric count labels are rendered to the right of each circle for direct value reading without requiring axis lookups. Tooltips display the director's name and exact title count on hover. The chart responds fully to cross-filtering, dynamically re-ranking directors based on the active filters.

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5. Interaction Techniques Applied

The dashboard is highly interactive, implementing the concept of linked views through a central state manager that coordinates all seven visualizations simultaneously. The following interaction techniques are implemented:

- **Cross-Filtering (Linked Views):** Selecting any element in one chart — such as clicking a country on the map or a bar in the ratings chart — instantly filters and updates all other visualizations in real-time.
- **Brush & Zoom on Timeline:** A brushable selection on the Content Growth Timeline allows users to select a specific year range, zooming focus across all charts to that time period.
- **Map Click-to-Filter:** Clicking any country on the Choropleth Map filters the entire dashboard to show only content produced in that country.
- **Treemap Drill-Down:** The Genre Treemap supports click-to-zoom, allowing users to drill into genre subcategories with breadcrumb navigation for easy backtracking.
- **Rating Bar Click:** Clicking any bar in the Content Ratings chart filters all other charts to show only titles with that specific rating.
- **Sidebar Filters:** Dropdown menus and toggle buttons in the sidebar allow filtering by Content Type (Movies / TV Shows), Country, Rating, and Genre independently.

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- **Smooth D3 Transitions (Object Constancy):** All charts animate smoothly when filters change, maintaining visual continuity so users can track how data shifts.
- **Radial Arc Animation:** The Seasonality chart animates arc lengths on every filter update, visually reinforcing changes in monthly content patterns.
- **Dynamic Director Re-Ranking:** The Top 10 Directors lollipop chart re-ranks and re-animates directors based on the active filter context.
- **Rich Tooltips:** Hovering over any data element across all charts displays a detailed tooltip showing exact values and relevant metadata.
- **Reset All Filters:** A single Reset button clears all active filters and restores the full dataset view instantly.

6. Challenges encountered and solutions applied

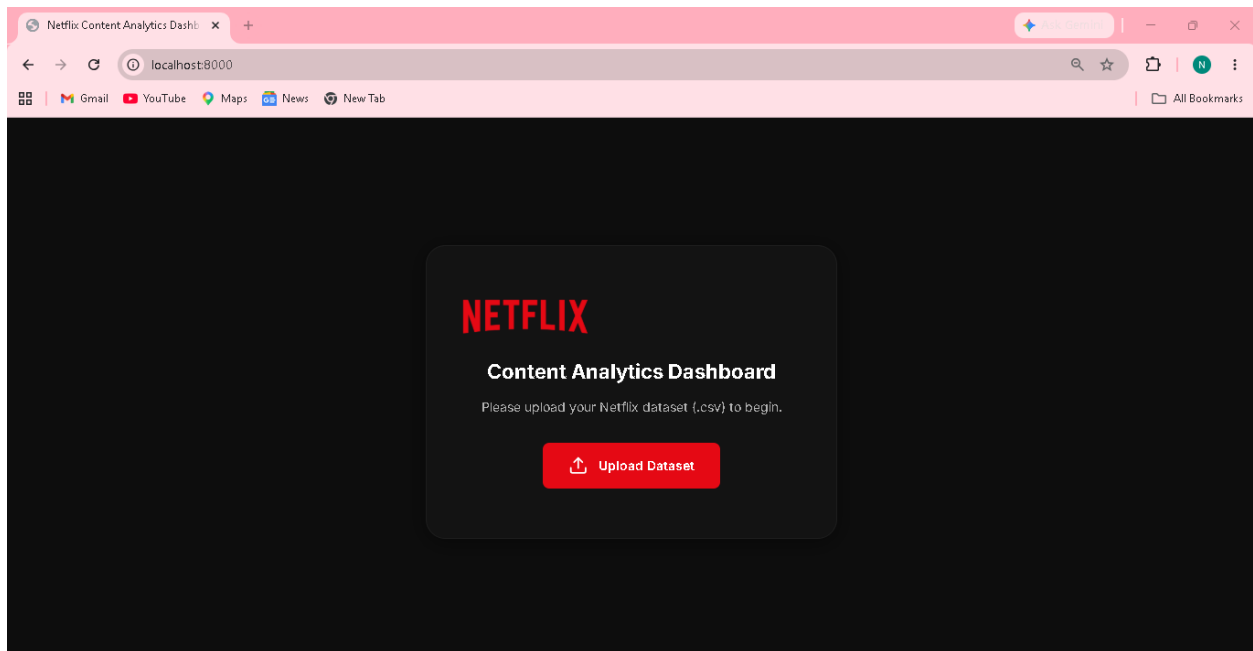
#	Challenge	Solution Applied
1	CORS Restriction when loading CSV	Served the project using a local Python HTTP server (python -m http.server 8000) to bypass CORS restrictions.
2	Dirty & Inconsistent Dataset	Developed a dedicated data loader to clean, normalize, and process data before visualization.
3	Cross-Filter State Synchronization	Implemented a centralized state manager to ensure consistent updates across all charts.
4	Choropleth Country Name Mismatch	Added a country name mapping and normalization layer to align dataset and geographic data.
5	Treemap Zoom & Breadcrumb Navigation	Maintained an internal zoom stack to manage drill-down navigation and filter updates.
6	Radial Chart Smooth Transitions	Used D3's attrTween() with stored arc states for smooth animated transitions.

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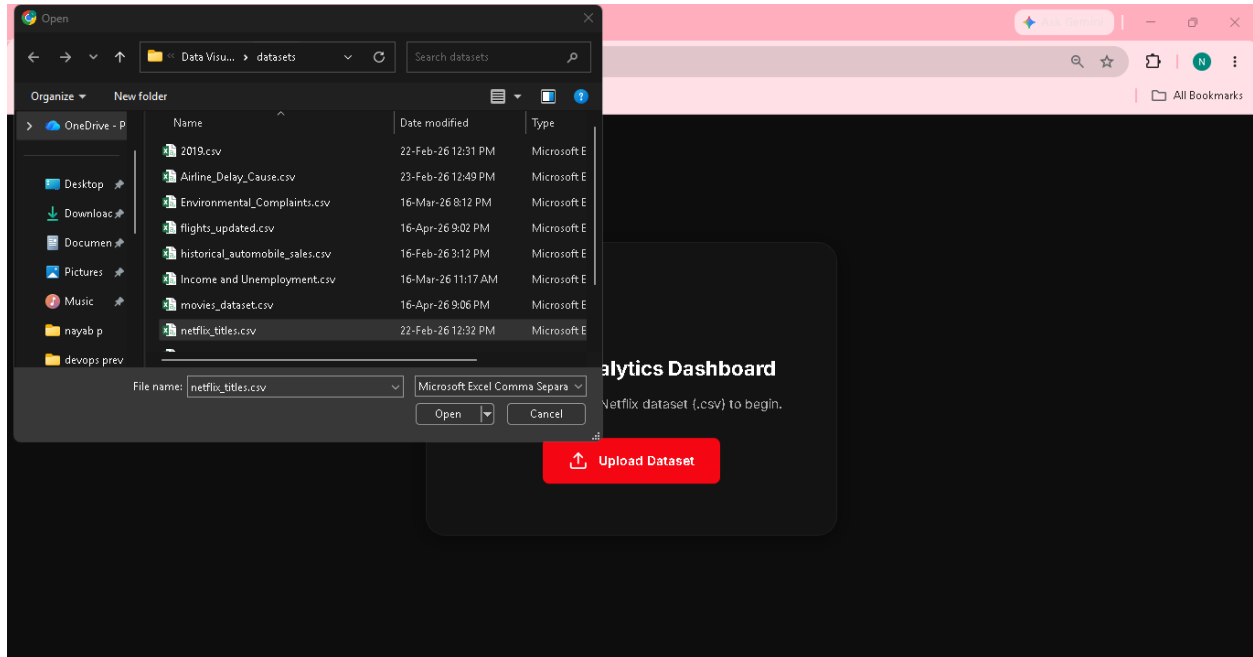
7	Responsive Layout Across Screen Sizes	Applied dynamic resizing and chart re-rendering using resize event handling.
8	Missing Director Information	Filtered out null, empty, and unknown director values before ranking calculations.
9	Missing Seasonality Data	Excluded records with invalid or missing dates during monthly aggregation.
10	Theme Toggle Redrawing Charts	Forced all charts to refresh and reapply updated CSS variables after theme changes.

Dashboard Output:

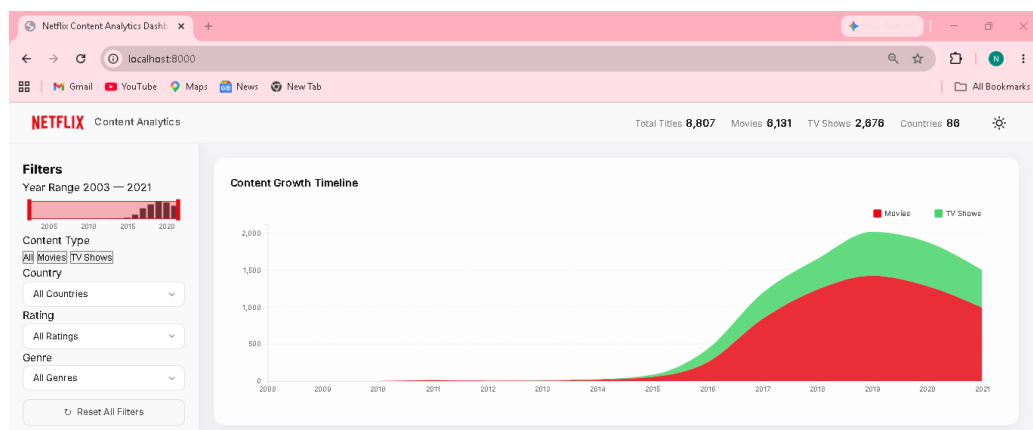
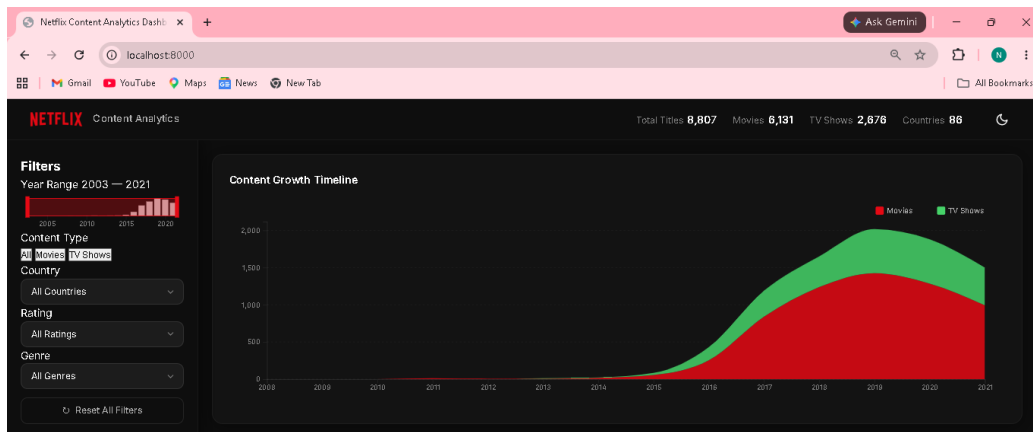
— Upload the CSV dataset



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—Implemented a toggle button for improved visualization



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— Added filters for Year Range, Content Type, Country, Rating, and Genre



— Developed the full dashboard view for comprehensive data analysis

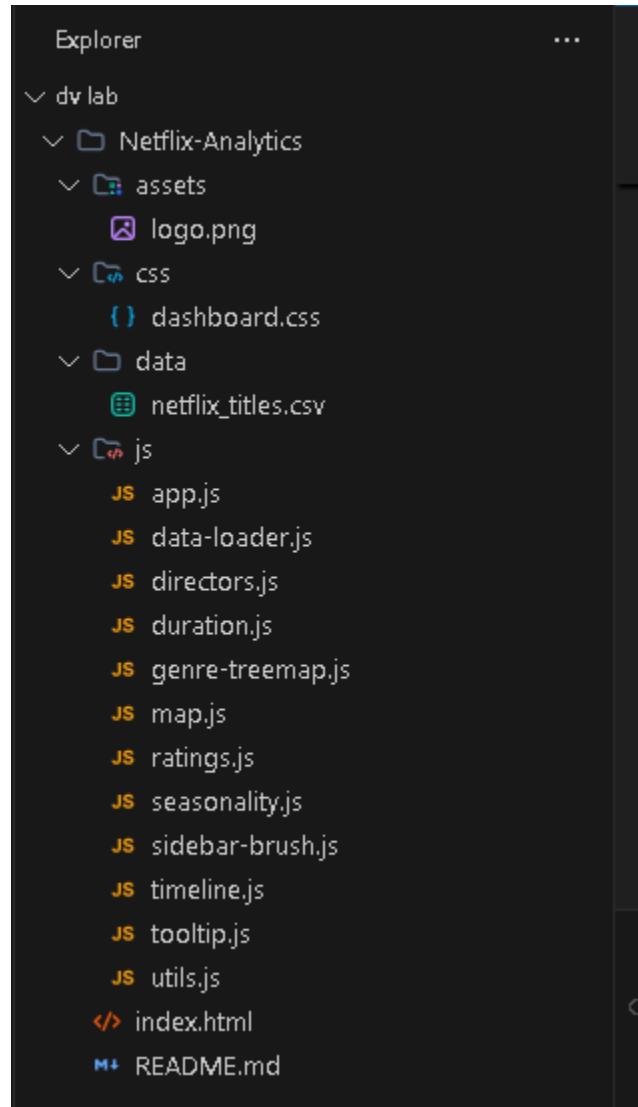


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`python -m http.server 8000`

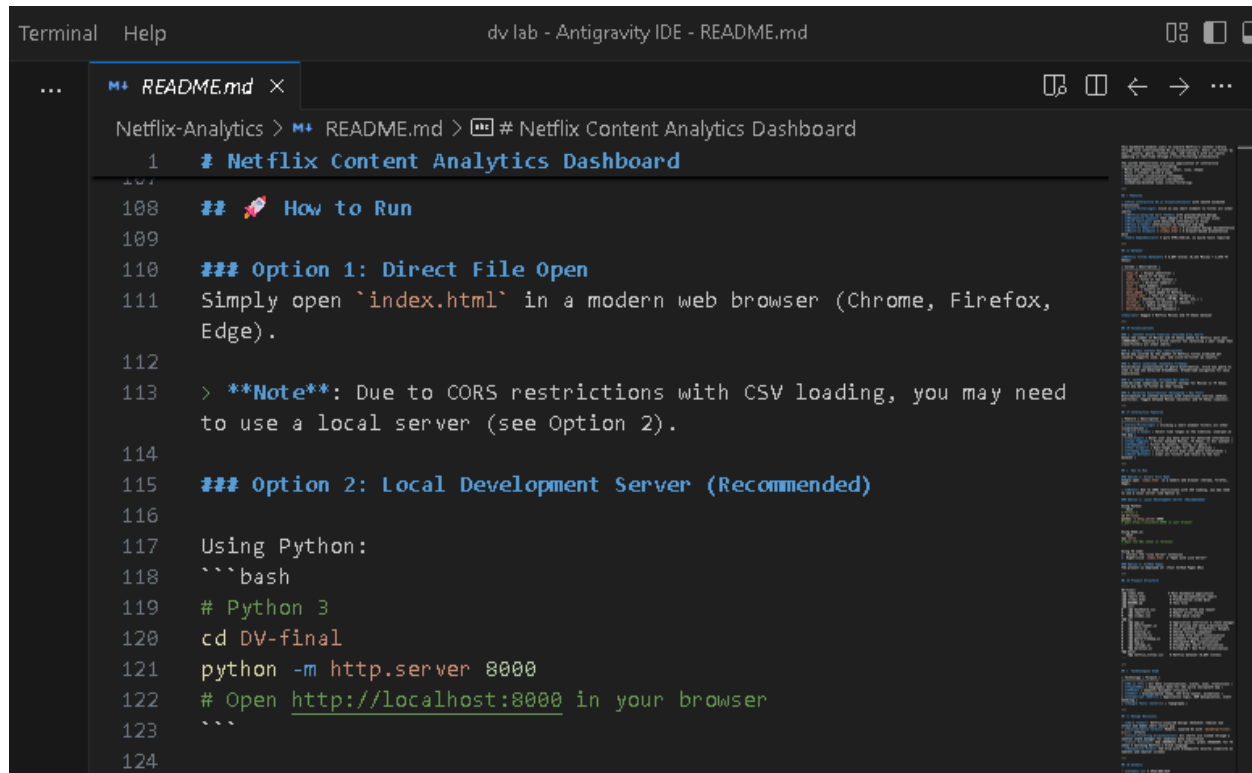
<http://localhost:8000>

Code Directory:



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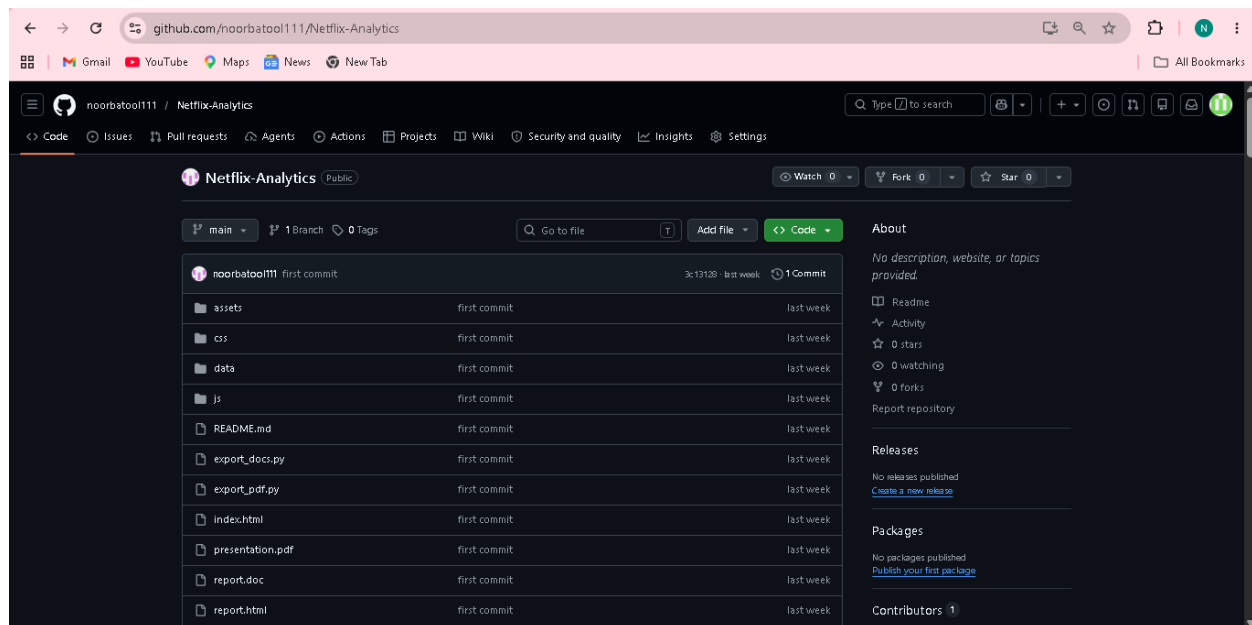
README file explaining how to run the application:



The screenshot shows a code editor window titled "dv lab - Antigravity IDE - README.md". The file content is as follows:

```
Netflix-Analytics > README.md > # Netflix Content Analytics Dashboard
1  # Netflix Content Analytics Dashboard
108 ## 🚀 How to Run
109
110 ### Option 1: Direct File Open
111 Simply open `index.html` in a modern web browser (Chrome, Firefox,
112 Edge).
113 > **Note**: Due to CORS restrictions with CSV loading, you may need
114 to use a local server (see Option 2).
115
116 ### Option 2: Local Development Server (Recommended)
117
118 Using Python:
119 ```bash
120 # Python 3
121 cd DV-final
122 python -m http.server 8000
123 # Open http://localhost:8000 in your browser
124 ```
```

Application hosted on Github:



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The screenshot shows the GitHub repository page for 'Netflix Content Analytics Dashboard' by noorbatool111. The page is dark-themed and features a README section with a title, description, and a table of contents. The table of contents includes links to Overview, Features, Dataset, Visualizations, Interactive Features, How to Run, Project Structure, Technologies Used, Design Decisions, and Authors. On the right side, there is a 'Languages' section showing the project's tech stack: JavaScript (59.0%), HTML (17.1%), CSS (16.2%), and Python (7.7%). Below this, there are 'Suggested workflows' for SLSA Generic generator, Node.js, and Webpack, each with a 'Configure' button. The browser's address bar shows the URL 'github.com/noorbatool111/Netflix-Analytics'.

github.com/noorbatool111/Netflix-Analytics

README

Netflix Content Analytics Dashboard

An interactive, web-based data visualization system built with D3.js that provides meaningful insights from the Netflix Titles dataset (8,800+ titles). This project was developed as a semester project for the Data Visualization course.

D3.js v7 License MIT Status Complete

Table of Contents

- Overview
- Features
- Dataset
- Visualizations
- Interactive Features
- How to Run
- Project Structure
- Technologies Used
- Design Decisions
- Authors

Languages

JavaScript 59.0% HTML 17.1% CSS 16.2% Python 7.7%

Suggested workflows

Based on your tech stack

- SLSA Generic generator: Generate SLSA provenance for your existing release workflows. [Configure](#)
- Node.js: Build and test a Node.js project with npm. [Configure](#)
- Webpack: Build a Node.js project with npm and webpack. [Configure](#)

[More workflows](#) [Dismiss suggestions](#)

The screenshot shows the 'Design Decisions' and 'Authors' sections of the GitHub repository page. The 'Design Decisions' section lists several key design choices: Dark Theme, Glassmorphism Cards, Cross-Filtering Architecture, Color Palette, and Responsive Grid. The 'Authors' section lists two students: Student 1 (SP23-BDS-039) and Student 2 (SP23-BDS-040). The 'License' section states that the project is created for educational purposes as part of the Data Visualization course. The browser's address bar shows the URL 'github.com/noorbatool111/Netflix-Analytics'.

github.com/noorbatool111/Netflix-Analytics

README

Design Decisions

- Dark Theme: Netflix-inspired design (#141414) reduces eye strain and makes chart colors pop
- Glassmorphism Cards: Modern, layered UI with `backdrop-filter: blur()` effects
- Cross-Filtering Architecture: All charts are linked through a central state manager for seamless data exploration
- Color Palette: Red (#E50914) for movies, green (#46d369) for TV shows — matching Netflix's brand language
- Responsive Grid: CSS Grid with breakpoints ensures usability on tablets and smaller screens

Authors

- Student 1 — SP23-BDS-039
- Student 2 — SP23-BDS-040

Data Visualization — Spring 2026

License

This project is created for educational purposes as part of the Data Visualization course, "Netflix-Analytics" "# Netflix-Analytics"

URL: <https://github.com/noorbatool111/Netflix-Analytics>

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Conclusion

The **Netflix Content Analytics Dashboard** successfully demonstrates the practical application of advanced data visualization techniques using **D3.js** to transform raw Netflix data into meaningful and interactive insights. Through comprehensive data preprocessing, exploratory analysis, and the implementation of seven linked visualizations, the project effectively reveals patterns related to content growth, geographic distribution, genres, ratings, duration, seasonality, and director contributions.

The integration of interactive features such as cross-filtering, brush-and-zoom, drill-down navigation, dynamic tooltips, and real-time updates enhances user exploration and understanding of the dataset. The dashboard not only provides valuable insights into Netflix's content library but also showcases how visualization can support data-driven decision-making and storytelling.

Overall, this project fulfills its objectives by combining analytical accuracy, interactive design, and effective visual communication, making it a strong demonstration of modern data visualization principles and techniques.